

Technical Note: Installing a Compatible xgboost Version for This Software

1. Overview

This software uses the R packages **caret** and **xgboost**.

The XGBoost model is trained via:

```
caret::train(method = "xgbTree", ...)
```

On some systems, especially on **Windows** with recent versions of **xgboost** or missing build tools, the training of the XGBoost model can fail. This note explains:

- The typical **error messages** related to **xgboost**;
- The **cause** of the problem (version and toolchain issues);
- The **steps required** to install a compatible version of **xgboost** so that the software runs correctly.

The rest of the software (Random Forest, SVM, Elastic-Net, MGM, Bayesian Network, Autoencoder) is **not affected** by this specific problem.

In this project, the recommended and tested version of **xgboost** is:

xgboost 1.7.6.1

All instructions below assume this version number.

2. Symptoms

When the XGBoost component is executed, you may see messages similar to:

```
Error in modelFit$xNames <- colnames(x) :  
  ALTLIST classes must provide a Set_elt method [class: XGBAltrepPointerClass, pkg: xgboost]
```

followed by a caret error such as:

```
Something is wrong; all the Accuracy metric values are missing:
```

Accuracy	Kappa
Min. : NA	Min. : NA
1st Qu.: NA	1st Qu.: NA
Median : NA	Median : NA
Mean : NaN	Mean : NaN
3rd Qu.: NA	3rd Qu.: NA
Max. : NA	Max. : NA
NA's : 8	NA's : 8

```
Error: Stopping
```

```
In addition: There were missing values in resampled performance measures.
```

When trying to install or change the version of **xgboost**, you may also see messages such as:

```
Could not find tools necessary to compile a package  
Call `pkgbuild::check_build_tools(debug = TRUE)` to diagnose the problem.
```

or:

```
'make' not found
```

3. Technical Cause

There are two main causes behind these errors:

1. Compatibility between `caret` and `xgboost`

- `caret::train(method = "xgbTree")` relies on an interface to `xgboost` that expects certain internal structures.
- Some **newer versions** of `xgboost` (especially on Windows) use implementation details (e.g. ALTREP-based internal classes) that are not fully aligned with the `xgbTree` model definition in `caret`.
- As a result, the assignment of variable names inside the `caret` model fit fails, leading to:

```
Error in modelFit$xNames <- colnames(x) :  
  ALTLIST classes must provide a Set_elt method [...]
```

- When every resampled model fit fails, `caret` ends up with **all NA performance metrics** and stops with:

```
Something is wrong; all the Accuracy metric values are missing
```

2. Missing compilation tools on Windows

- When installing a specific version of `xgboost` from source (using `remotes::install_version()`), R needs **build tools** (compiler, `make`, etc.).
- On Windows, these are provided by **Rtools**.
- If Rtools is not installed or not correctly configured, installation fails with:

```
Could not find tools necessary to compile a package  
'make' not found
```

To make this software work correctly with XGBoost, you must:

- Ensure a **working toolchain** (Rtools) is available on Windows;
- Install the specific tested version **`xgboost 1.7.6.1`**.

4. Environment Requirements (Windows)

Before installing `xgboost`, please ensure:

- You are using a supported version of **R 4.x**;
- The corresponding **Rtools** version is installed (e.g. Rtools 4.4 for R 4.4);
- Your PATH allows R to find `make` and other build tools.

4.1 Install and Configure Rtools

1. Download Rtools

- Go to the official CRAN Rtools page (e.g. search “CRAN Rtools” in your browser).
- Download the Rtools installer appropriate for your version of R (for example, Rtools 4.4 for R 4.4).

2. Install Rtools

- Run the downloaded `.exe` file (e.g. `rtools44-xxxx-xxxx.exe`).
- Accept the default installation path (e.g. `C:\rtools44`).
- Keep the option to add Rtools to the system PATH enabled, if offered.
- Complete the installation.

3. Restart R / RStudio

- After installation, close and reopen R or RStudio.

4. Verify build tools in R

In R:

```
install.packages("pkgbuild")           # if not already installed
library(pkgbuild)

pkgbuild::check_build_tools(debug = TRUE)
Sys.which("make")
```

- `pkgbuild::check_build_tools()` should **not** report an error.
- `Sys.which("make")` should return a non-empty path, for example:

```
"C:\\rtools44\\usr\\bin\\make.exe"
```

If `make` is still not found, verify that Rtools is correctly installed and added to the PATH, then restart R and try again.

5. Installing xgboost Version 1.7.6.1

Once the build tools are working, you can install the **fixed version** of `xgboost` that is known to work with this software and with `caret::xgbTree`.

5.1 Remove any existing xgboost installation

In R:

```
remove.packages("xgboost")
```

Restart R / RStudio after this step if you want a completely clean environment.

5.2 Install remotes (if necessary)

```
install.packages("remotes")
library(remotes)
```

5.3 Install xgboost 1.7.6.1

```
remotes::install_version(
  "xgboost",
  version = "1.7.6.1",
  repos   = "https://cran.r-project.org"
)
```

5.4 Verify the installed version

```
packageVersion("xgboost")
# should print '1.7.6.1'
```

If the printed version is 1.7.6.1, the `xgboost` installation is correct for this software.

6. Rerunning the Software

After the above steps:

1. Ensure that:
 - `pkgbuild::check_build_tools(debug = TRUE)` reports success;
 - `Sys.which("make")` returns a valid path;
 - `packageVersion("xgboost")` is exactly 1.7.6.1.
2. Restart R / RStudio.
3. Rerun your analysis or script that uses:

```
caret::train(method = "xgbTree", ...)
```

If the environment is correctly configured:

- The XGBoost model should train successfully inside `caret` (no more `ALTLIST` errors).
 - Cross-validation metrics (Accuracy, F1-score, Matthews correlation coefficient, etc.) should be computed without missing values.
 - The full pipeline of models in the software, including XGBoost, will execute as intended.
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7. Information to Provide for Support

If, after following this note, you still experience errors related to `xgboost`, please provide the following information when contacting the development team:

```
sessionInfo()           # full R session information
packageVersion("xgboost") # installed xgboost version (should be 1.7.6.1)
packageVersion("caret")  # caret version
pkgbuild::check_build_tools(TRUE) # result of the build tools check
Sys.which("make")         # path to 'make'
```

Including these details will help diagnose any remaining environment-specific issues quickly and accurately.